

2. Algebra

Ref	Content	Notes
2.1	The basic processes of algebra	Knowledge and use of basic skills in manipulative algebra including use of the associative, commutative and distributive laws, are expected
2.2	Definition of a function	Notation $f(x)$ will be used, e.g. $f(x) = x^2 - 9$
2.3	Domain and range of a function	Domain may be expressed as, for example, $x > 2$, or 'for all x , except $x = 0$ ' and range may be expressed as $f(x) > -1$
2.4	Composite functions	The result of two or more functions, say f and g , acting in succession. $fg(x)$ is g followed by f
2.5	Inverse functions	The inverse function of f is written f^{-1} Domains will be chosen for f to make f one-one
2.6	Expanding brackets and collecting like terms	Expand and simplify $(y^2 - 2y + 3)(2y - 1) - 2(y^3 - 3y^2 + 4y - 2)$
2.7	Expand $(a + b)^n$ for positive integer n	Expand and simplify $(5x + 2)^3$ Use Pascal's triangle to work out the coefficient of x^3 in the expansion of $(3 + 2x)^5$
2.8	Factorising	Factorise fully $(2x + 3)^2 - (2x - 5)^2$ Factorise $15x^2 - 34xy - 16y^2$ Factorise fully $x^4 - 25x^2$
2.9	Manipulation of rational expressions: Use of $+$ $-$ $\times$ $\div$ for algebraic fractions with denominators being numeric, linear or quadratic	Simplify $\frac{5}{x+2} - \frac{3}{2x-1}$ Simplify $\frac{x^3 + 2x^2 + x}{x^2 + x}$ Simplify $\frac{5x^2 - 14x - 3}{4x^2 - 25} \div \frac{x-3}{4x^2 + 10x}$
2.10	Use and manipulation of formulae and expressions	Rearrange $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ to make v the subject

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2.11	Use of the factor theorem for rational values of the variable for polynomials	Factorise $x^3 - 2x^2 - 5x + 6$ Show that $2x - 3$ is a factor of $2x^3 - x^2 - 7x + 6$ Solve $x^3 + x^2 - 10x + 8 = 0$ Show that $x - 7$ is a factor of $x^5 - 7x^4 - x + 7$
2.12	Completing the square	Work out the values of a, b and c such that $2x^2 + 6x + 7 \equiv a(x + b)^2 + c$
2.13	Drawing and sketching of functions Interpretation of graphs	Graphs could be linear, quadratic, exponential and restricted to no more than 3 domains Exponential graphs will be of the form $y = ab^x$ and $y = ab^{-x}$, where a and b are rational numbers Sketch the graph of $y = x^2 - 5x + 6$ Label clearly any points of the intersection with the axes A function f is defined as $f(x) = \begin{cases} x^2 & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \\ 3 - x & 2 \leq x < 3 \end{cases}$ Draw the graph of $y = f(x)$ on the grid below for values of x from 0 to 3 Given a sketch of $y = ab^{-x}$, and two points, work out the values of a and b
2.14	Solution of linear and quadratic equations	Solutions of quadratics to include solution by factorisation, by graph, by completing the square or by formula Problems will be set in a variety of contexts, which result in the solution of linear or quadratic equations
2.15	Algebraic and graphical solution of simultaneous equations in two unknowns, where the equations could both be linear or one linear and one second order	Solve $4x - 3y = 0$ and $6x + 15y = 13$ Solve $y = x + 2$ and $y^2 = 4x + 5$ Solve $y = x^2$ and $y - 5x = 6$ Solve $xy = 8$ and $x + y = 6$

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2.16	Algebraic solution of linear equations in three unknowns	Solve $2x - 5y + 4z = 22$ $x + y + 2z = 4$ $x - y - 6z = -4$
2.17	Solution of linear and quadratic inequalities	Solve $5(x - 7) > 2(x + 1)$ Solve $x^2 < 9$ Solve $2x^2 + 5x \leq 3$
2.18	Index laws, including fractional and negative indices and the solution of equations	Express as a single power of x $\sqrt{x^{\frac{1}{2}} \times x^{\frac{7}{2}}}$ Express as a single power of x $\sqrt{\frac{x^{\frac{3}{2}} \times x^{\frac{7}{2}}}{x^2}}$ Solve $x^{-\frac{1}{2}} = 3$ Solve $\sqrt{x} - \frac{10}{\sqrt{x}} = 3 \quad x > 0$
2.19	Algebraic proof	Prove $(n + 5)^2 - (n + 3)^2$ is divisible by 4 for any integer value of n
2.20	Using n th terms of sequences Limiting value of a sequence as $n \rightarrow \infty$	Work out the difference between the 16 th and 6 th terms of the sequence with n th term $\frac{2n}{n + 4}$ Write down the limiting value of $\frac{2n}{n + 4}$ as $n \rightarrow \infty$
2.21	n th terms of linear sequences	A linear sequence starts 180 176 172 ... By using the n th term, work out which term has value -1000 Work out the n th term of the linear sequence $r + s \quad r + 3s \quad r + 5s \quad \dots$
2.22	n th terms of quadratic sequences	Work out the n th term of the quadratic sequence 10 16 18 16 ... Which term has the value 0?